



Overview of Advanced Interpolation Methods

MAT 425

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OUTLINE

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Lagrange Interpolation

The first 1D interpolation method discussed was **Lagrange Interpolation**. It constructs a polynomial from a given set of data points to estimate values between them.

Given $n + 1$ points:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

The interpolation polynomial is:

$$P(x) = \sum_{j=0}^n y_j \cdot L_j(x)$$

where the basis polynomial is:

$$L_j(x) = \prod_{\substack{0 \leq m \leq n \\ m \neq j}} \frac{x - x_m}{x_j - x_m}$$

Newton's Divided Difference

Newton's Divided Difference is a 1D interpolation method that uses a recursive table of divided differences to compute coefficients.

Given data points:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$
x_0	y_0			
x_1	y_1	$\frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_2	y_2	$\frac{f[x_2] - f[x_1]}{x_2 - x_1}$	$\frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$	
x_3	y_3	$\frac{f[x_3] - f[x_2]}{x_3 - x_2}$	$\frac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1}$	$\frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0}$

Cubic Spline Interpolation

Cubic spline interpolation fits a piecewise cubic polynomial between data points, ensuring smoothness by enforcing continuity of the first and second derivatives.

Conditions for cubic splines:

1. $S(x)$ is a cubic polynomial $S_j(x)$ on $[x_j, x_{j+1}]$ for $j = 0, \dots, n-1$.
2. $S_j(x_j) = f(x_j)$, $S_j(x_{j+1}) = f(x_{j+1})$.
3. $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$ function continuity.
4. $S'_j(x_{j+1}) = S'_{j+1}(x_{j+1})$ first derivative continuity.
5. $S''_j(x_{j+1}) = S''_{j+1}(x_{j+1})$ second derivative continuity.
6. Boundary conditions (choose one):
 - 6.1 $S''(x_0) = S''(x_n) = 0$ natural (free) BC.
 - 6.2 $S'(x_0)$ and $S'(x_n)$ specified clamped BC.

Overview of Paper 1

- Goal: Improve the angular resolution of HARDI signals without requiring longer scan times.
- Uses measured directions to predict unmeasured ones via local signal similarity.
- Synthesizes signals at predicted directions to reconstruct more accurate ODFs.

Predicting Directions

- Directions grouped into triangles based on correlation.
- A new direction u_c is predicted:

$$u_c = \sum_{k=1}^K w_k g_{jk}, \quad \sum w_k = 1, \quad w_k \geq 0$$

- Signal at that direction:

$$S(u_c) = \sum_{k=1}^K w_k S(g_{jk})$$

RBF Interpolation

- Gaussian radial basis function applied over sphere:

$$S(v_j) = \sum_{g \in D_C} c_g \cdot \exp\left(-\alpha \|v_j - g\|^2\right)$$

- Solve via:

$$(A + \lambda I)c = b$$

Paper 1: Validation

Quantitative Evaluation

- Three datasets were compared:
 - f_{64} Full scan with 64 directions
 - f_{32} Subset with 32 directions
 - f_{cm} 32 directions plus interpolated directions
- Relative L2 errors used as metric:

$$RE_{32} = \frac{\|\tilde{f}_{64} - \tilde{f}_{32}\|}{\|\tilde{f}_{64}\|}, \quad RE_{cm} = \frac{\|\tilde{f}_{64} - \tilde{f}_{cm}\|}{\|\tilde{f}_{64}\|}$$

- f_{cm} shows much lower reconstruction error than f_{32} (Table 1, 2).

Paper 1: Validation

Fiber Crossing & Structure

- Crossing fibers better preserved in interpolated data (fig. 5).
- Ratio of second fiber directions for f_{cm} closer to f_{64} than for f_{32} (Table 3).
- Qualitative images show smoother and more accurate voxel-level SDFs (fig. 4).

Paper 2: Q-ball Imaging

Goal

The main goal of paper 2 is to propose a solution for Q-ball imaging reconstruction of the orientation distribution function (ODF).

- ODF: $G[S](\mathbf{u}) = \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) S(\mathbf{w}) d\mathbf{w}$
- Estimate the function on a q-ball using spherical harmonics

Definition: A spherical harmonic of degree n denoted by $Y_n(\xi)$, is a harmonic homogeneous polynomial of degree n restricted to the unit 2-sphere.

$Y_n(\xi) = P_n|_{\mathbb{S}^2}$ which means

$$Y_n : \mathbb{S}^2 \rightarrow \mathbb{R}, \text{ such that } Y_n(\xi) = P_n(\xi) \quad \forall \xi \in \mathbb{S}^2$$

Paper 2: Q-ball Imaging

Theory

- Approximated signal at each of the gradient directions:

$$S(\theta_i, \phi_i) = \sum_{j=1}^R c_j Y_j(\theta_i, \phi_i)$$

- They use $R = \frac{(l+1)(l+2)}{2}$
- This can be written as: $S = BC$
- Need to solve for coefficients:

$$c_j = \int_{\Omega} S(\theta, \phi) Y_j(\theta, \phi) d\Omega$$

Paper 2: Q-ball Imaging

Theory

- They add local regularization in order to use a high order estimation without over-modeling small perturbations with Laplace-Beltrami Operator
- Thus they define a measure, E , of the deviation from smoothness of a function f on the unit sphere as:

$$E(f) = \int_{\Omega} (\Delta_b f)^2 d\Omega$$

- Since the SH basis is orthonormal this can be written as:

$$E(f) = \int_{\Omega} \Delta_b \left(\sum_p c_p Y_p \right) \Delta_b \left(\sum_q c_q Y_q \right) d\Omega =$$

$$\sum_{j=1}^R c_j^2 l_j^2 (l_j + 1)^2 = C^T L C$$

Paper 2: Q-ball Imaging

Theory

- Matrix form: $M(C) = (S - BC)^T(S - BC) + \lambda C^T L C$
- $C = (B^T B + \lambda L)^{-1} B^T S$
- From this the SH coefficient of the signal on the q-ball for any (θ, ϕ) can be recovered as:

$$S(\theta_i, \phi_i) = \sum_{j=1}^R c_j Y_j(\theta_i, \phi_i)$$

Paper 2: Q-ball Imaging

Theory

- Replacing the signal by the SH series approximation:

$$G[S](\mathbf{u}) = \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) S(\mathbf{w}) d\mathbf{w} = \sum_{j=1}^R c_j \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) Y_j(\mathbf{w}) d\mathbf{w}.$$

- Funk-Hecke Theorem:** Let $f(t)$ be continuous on $[-1, 1]$ and H_l any SH of order l in C^3 . Then, given a unit vector $\mathbf{u}$

$$\int_{|\mathbf{w}|=1} f(\mathbf{u}^T \mathbf{w}) H_l(\mathbf{w}) d\mathbf{w} = \lambda(l) H_l(\mathbf{u}),$$

where

$$\lambda(l) = 2\pi \int_{-1}^1 P_l(t) f(t) dt$$

where P_l is the Legendre polynomial of degree l .

Paper 2: Q-ball Imaging

Theory

- Using Gaussian to approximate Dirac Delta Function:

$$\int_{|w|=1} \delta(\mathbf{u}^T \mathbf{w}) Y_j(\mathbf{w}) d\mathbf{w} = 2\pi P_l(0) Y_j(u),$$

where $P_l(0)$ is the Legendre Polynomial of degree l evaluated at 0.

- Using combinatorics techniques it can be obtained that when l is odd $P_l(0)$ is 0 and when l is even $P_l(0)$ is $(-1)^{\frac{l}{2}} \frac{1*3*5*\dots*(l-1)}{2*4*6*\dots*l}$. Now this can be applied to the ODF:

$$G[S](\mathbf{u}) = \sum_{j=1}^R 2\pi P_{l_j}(0) c_j Y_j(\mathbf{u})$$

MATLAB

Input and Output

Inputs

- $\text{gdir} \in \mathbb{R}^{240 \times 3}$
- $\text{rho4000} \in \mathbb{R}^{96 \times 96 \times 40 \times 240}$
- idx : Red direction indices; remaining are blue
- ii , ij , nth_sl : Selected voxel location (e.g., $\text{ii}=23$, $\text{ij}=12$, $\text{nth_sl}=24$).
- λ : Regularization parameter
- rbft : Radial basis function type (e.g. Gaussian)

Outputs

- estimated_bS : Interpolated signal at blue directions
- ODFtS , ODFvS : Orientation Distribution Functions

MATLAB

Algorithm Steps

1. **Load Data:** Import `gdirt`, `rho4000`
2. **Direction Split:** Divide into red (measured) and blue (to be estimated)
3. **Signal Extraction:**
 - `tS`: all directions
 - `vS`: red directions
 - `uS`: blue directions (ground truth)
4. **Green Directions:**
 - Use `get_cm()` to get `cmdirt`, `cmval`
5. **Prepare RBF Input:**
 - `rg_directions` = [`vgdirt`; `-vgdirt`; `cmdirt`; `-cmdirt`]
 - `rg_signal` = [`vS`; `vS`; `cmval`; `cmval`]

MATLAB

Algorithm Steps

6. Optimize ε :

- `ep_opt = LOOCV3Dmin(rg_directions, rg_signal, rbft)`

7. Interpolate:

`estimated_bS = RBFInterpolation(rg_directions, rg_signal, ugdirt, rbft, ep_opt,  $\lambda$ )`

8. ODF Reconstruction:

`ODF = mkODF([vgdirt; ugdirt], [vS; estimated_bS],  $\lambda$ )`

9. Visualization:

- Use `plotQBI()` to compare ODFs

MATLAB Images

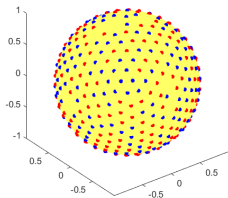


Figure: Q-ball

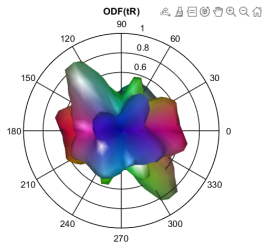


Figure: ODF from only red signal values

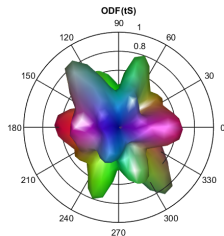


Figure: ODF from all signal values

Brain Tractography

Brain tractography is a post-processing technique used in diffusion MRI to reconstruct the probable pathways of white matter fiber tracts in the brain using Diffusion Spectrum Imaging (DSI) software.

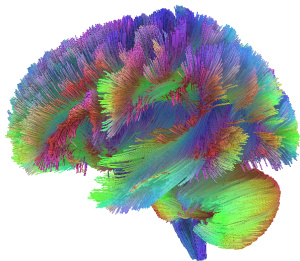


Figure: Lateral View of Brain Tractography

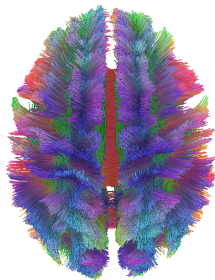


Figure: Superior View of Brain Tractography